

Research Methodology in CS (CSEG3060_3)

Lecture 7: Ethical Considerations in Research: informed consent, confidentiality, data fabrication/falsification, authorship norms, and conflict of interest

Unit 3: Ethical Conduct and Literature Studies • Mapped Course Outcome: CO3

Lecture 7: Ethical Considerations in Research

Informed Consent, Confidentiality, Data Fabrication/Falsification, Authorship Norms, and Conflict of Interest

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science)
Semester: 5 **Unit:** 3 — Ethical Conduct and Literature Studies **Lecture:** 7 of the unit **Mapped Course Outcome:** CO3 — Demonstrate an understanding of ethical practices, scholarly conduct, and responsible research behaviour.

1. Lecture Overview

Research, irrespective of discipline, is built upon a foundation of trust. The credibility of scientific knowledge depends not only on the rigour of methods used but also on the ethical integrity of the researchers who produce it. In computer science, where research increasingly involves human participants (HCI, user studies, AI fairness assessments), large-scale datasets, and multi-author collaborations, ethical awareness is no longer optional—it is a core competency.

This lecture provides a structured treatment of the five pillars of research ethics most frequently encountered by CS researchers:

1. Informed consent
2. Confidentiality and data protection
3. Data fabrication and falsification
4. Authorship norms
5. Conflict of interest

Each section includes definitions, regulatory references (such as the ICMR guidelines, the Belmont Report principles, and COPE/ICMJE authorship standards), illustrative examples, and computer science-specific applications.

2. Learning Objectives

By the end of this lecture, students should be able to:

- Define and apply the principle of informed consent in human-subjects research.
 - Explain the mechanisms used to preserve confidentiality and protect data.
 - Distinguish between data fabrication, falsification, and legitimate data transformation.
 - Identify the criteria for legitimate authorship and the consequences of honorary or ghost authorship.
 - Recognize, disclose, and manage conflicts of interest in research.
 - Connect each ethical principle to concrete computer science research scenarios.
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3. Key Concepts at a Glance

Ethical Domain	Core Principle	Primary Risk if Violated
Informed Consent	Voluntary, informed, capacitated agreement	Coercion, exploitation
Confidentiality	Privacy protection and controlled disclosure	Privacy breach, identity exposure
Data Fabrication/Falsification	Honest data collection and reporting	Loss of scientific validity
Authorship Norms	Credit for substantial intellectual contribution	Misattribution of credit
Conflict of Interest	Transparent disclosure of competing interests	Compromised objectivity

4. Detailed Explanations

4.1 Informed Consent

4.1.1 Definition

Informed consent is a process by which a participant voluntarily agrees to take part in a research study after being fully informed about its purpose, procedures, risks, benefits, alternatives, and

the handling of their data. It is rooted in the Belmont Report's principle of *respect for persons* and reaffirmed in Article 6 of the Declaration of Helsinki.

4.1.2 Components of Valid Informed Consent

A valid informed consent typically contains the following elements:

- **Disclosure:** The researcher must disclose the study's purpose, duration, procedures, foreseeable risks, expected benefits, and any compensation.
- **Understanding:** Information must be presented in language that the participant can comprehend, free from jargon or undue influence.
- **Voluntariness:** Participation must be free from coercion, undue pressure, or penalty for non-participation.
- **Capacity:** The participant must have the legal and cognitive capacity to consent. For minors or incapacitated individuals, surrogate or parental consent is required.
- **Comprehension and Documentation:** Usually recorded through a signed consent form, although in some digital contexts, click-through or recorded verbal consent is acceptable.

4.1.3 Special Considerations in Computer Science

- **Online experiments and crowdsourcing (e.g., Amazon Mechanical Turk, Prolific):** Consent must be obtained prior to task exposure. IP addresses and metadata may constitute identifiable information.
- **AI fairness and dataset research:** Consent issues arise when researchers use pre-existing datasets containing personal information. Secondary use of data without specific consent may violate the original agreement.
- **A/B testing in industry:** Many large tech firms conduct experiments on users without explicit consent. This raises ethical concerns regarding autonomy and transparency.

4.1.4 CS Research Example

A researcher studying the usability of a new authentication mechanism recruits 100 participants. Each participant must be told:

- That they may withdraw at any time.
- That their keystroke dynamics data will be collected.
- How the data will be stored and destroyed.
- Whom to contact for grievances.

Failure to disclose the keystroke logging constitutes an ethical violation, even if the protocol was institutionally approved.

4.2 Confidentiality

4.2.1 Definition

Confidentiality refers to the obligation of researchers to protect the private information of participants and prevent unauthorized access, use, or disclosure. It is distinct from anonymity: anonymized data cannot be linked back to an individual, while confidential data are linkable but protected by safeguards.

4.2.2 Principles of Data Protection

Confidentiality is operationalized through four key practices:

1. **Minimization:** Collect only the data necessary to answer the research question.
2. **Anonymization and Pseudonymization:** Remove or replace direct identifiers.
3. **Secure Storage:** Use encryption, access controls, and audit logs.
4. **Controlled Access:** Restrict data access to authorized personnel under binding agreements.

4.2.3 Regulatory Frameworks

- **India:** The Digital Personal Data Protection Act (DPDPA), 2023, governs processing of digital personal data.
- **European Union:** The General Data Protection Regulation (GDPR) sets strict consent and data-handling standards.
- **International:** The Health Insurance Portability and Accountability Act (HIPAA) governs medical data confidentiality.

4.2.4 Technical Mechanisms

Confidentiality can be reinforced using:

- **Encryption at rest and in transit:** AES-256 for storage, TLS 1.3 for transmission.
- **Differential Privacy:** Adding calibrated noise to datasets to limit re-identification risk.
- **Federated Learning:** Training models without transferring raw data to a central server.
- **Secure Multi-Party Computation:** Allowing joint computation without exposing individual inputs.

These mechanisms are particularly important in CS research that involves health informatics, social network analysis, or behavioural analytics.

4.3 Data Fabrication and Falsification

4.3.1 Definitions

The U.S. Office of Research Integrity (ORI) and most institutional policies classify research misconduct into three categories, commonly known as FFP:

- **Fabrication:** Making up data, results, or records and reporting them as if they were real.
- **Falsification:** Manipulating research materials, equipment, or processes, or changing, omitting, or selectively reporting data, such that the research is not accurately represented.
- **Plagiarism:** Appropriation of another person's ideas, processes, results, or words without giving appropriate credit (covered in the previous lecture).

4.3.2 The Spectrum from Acceptable to Unacceptable Practices

Not all data modifications are unethical. The following table illustrates legitimate versus illegitimate practices:

Practice	Ethical Status
Removing outliers with documented statistical justification	Generally acceptable
Applying anonymization or aggregation	Acceptable and often required
Cherry-picking results that support a hypothesis while omitting null results	Unethical (selective reporting)
Generating synthetic data to fill missing records without disclosure	Unethical fabrication
Re-running an experiment until $p < 0.05$ (p-hacking)	Unethical manipulation
Image manipulation in figures that alters content	Unethical falsification

4.3.3 Computer Science Specific Issues

- **Image manipulation:** Western blots and microscopy images in computational biology papers may be manipulated using software like Adobe Photoshop. Journals now require raw image data.

- **Algorithm benchmarking:** Cherry-picking benchmarks, omitting unfavourable datasets, or tuning hyperparameters on test sets constitutes a form of falsification.
- **Reproducibility crises:** Studies have shown that many published ML results cannot be replicated, partly due to undisclosed data preprocessing or selective reporting.

4.3.4 Notable Case Studies

- **The Hwang Woo-suk case (2004–2005):** South Korean researcher fabricated data in stem-cell research, leading to retraction of high-profile papers and criminal conviction.
- **The Jan Hendrik Schön case (2002):** A physicist at Bell Labs fabricated data across multiple semiconductor studies.

In CS, the **Heidi Mitchell case (2023)** and numerous retractions tracked on Retraction Watch highlight how benchmark manipulation and undisclosed hyperparameter tuning can lead to formal investigations.

4.4 Authorship Norms

4.4.1 Why Authorship Matters

Authorship is the primary mechanism through which researchers receive credit for intellectual contributions. Misattribution distorts merit, undermines accountability, and affects career progression, funding decisions, and public trust.

4.4.2 International Standards

The **ICMJE (International Committee of Medical Journal Editors)** and **COPE (Committee on Publication Ethics)** have established widely accepted authorship criteria. Under the ICMJE guidelines, authorship requires:

1. Substantial contributions to conception, design, data acquisition, analysis, or interpretation.
2. Drafting or critical revision for important intellectual content.
3. Final approval of the version to be published.
4. Accountability for all aspects of the work.

All four conditions must be met. Contributors who do not meet all four may be acknowledged but not listed as authors.

4.4.3 Common Authorship Misconducts

Misconduct	Description
Gift Authorship	Listing senior colleagues without their meaningful contribution.
Guest Authorship	Adding a well-known researcher to enhance credibility.
Ghost Authorship	Omitting a contributor (e.g., a medical writer or junior researcher) who made substantive contributions.
Coercive Authorship	Senior authors pressuring juniors to include unrelated contributors.
Mutual Benefit Authorship	Researchers agreeing to add each other across papers without genuine contribution.

4.4.4 Computer Science Specific Concerns

- **Software contributions:** Traditional authorship criteria were designed for individuals. Software is often produced collaboratively. Researchers should follow CRediT (Contributor Roles Taxonomy) to specify roles.
- **Large-scale collaborative ML projects:** Papers in conferences like NeurIPS may have 30+ authors. Contribution transparency becomes critical.
- **Advisor-student dynamics:** Students must be listed as first author when they are the primary contributors. This is a recurring source of conflict.

4.4.5 CRediT Taxonomy

The CRediT system defines 14 contributor roles, including:

- Conceptualization
- Methodology
- Software
- Validation
- Formal analysis
- Investigation
- Data curation
- Writing – original draft preparation
- Writing – review and editing
- Visualization

- Supervision
- Project administration
- Funding acquisition
- Resources

Using CRediT promotes transparency in multi-author CS research.

4.5 Conflict of Interest (COI)

4.5.1 Definition

A conflict of interest exists when a researcher's personal, financial, professional, or institutional interests could compromise—or appear to compromise—the objectivity, integrity, or independence of the research.

4.5.2 Types of Conflicts

1. **Financial COIs:** Stock holdings, consulting fees, royalties, or honoraria from entities that could benefit from the research outcome.
2. **Personal COIs:** Close relationships with individuals or organizations that could bias the research.
3. **Institutional COIs:** Institutional pressures, such as funding from a company whose product is being evaluated.
4. **Intellectual COIs:** Strongly held theoretical commitments that may bias interpretation of data.
5. **Political or Ideological COIs:** Particularly relevant in AI ethics, algorithmic fairness, and policy research.

4.5.3 Disclosure Requirements

Most journals and conferences require a mandatory COI disclosure. IEEE, ACM, NeurIPS, and ICML all require authors to declare:

- Funding sources.
- Employment relationships.
- Equity interests.
- Advisory board memberships.
- Any other relevant relationships within the past three years.

Failure to disclose is itself a violation, regardless of whether the research was actually biased.

4.5.4 Managing Conflicts of Interest

Effective management strategies include:

- **Disclosure:** Full transparency to journals, funders, and ethics committees.
- **Recusal:** Removing oneself from review or decision-making processes.
- **Independent Oversight:** Having an independent statistician or auditor verify results.
- **Blinding:** Using double-blind review (where feasible).
- **Recombination of Research Teams:** Including members without the conflict to balance perspectives.

4.5.5 CS Research Examples

- **Industry-sponsored AI research:** A researcher evaluating the safety of a self-driving system funded by the manufacturer has a clear financial COI.
 - **Open-source software analysis:** A developer who maintains a competing library may have a bias when benchmarking performance.
 - **Algorithmic auditing:** Researchers auditing a hiring algorithm from a company they once consulted for must disclose prior relationships.
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5. Cross-Cutting Themes

5.1 The Interdependence of Ethical Principles

The five ethical domains are not isolated. A single study may require:

- Informed consent from participants.
- Confidentiality protection for the data collected.
- Honest reporting without fabrication or falsification.
- Accurate authorship attribution.
- Transparent disclosure of any conflicts.

Failure in one domain often cascades into others. For example, falsification of data typically involves misleading co-authors, violating authorship norms and trust.

5.2 Institutional Mechanisms

Research ethics are enforced through:

- **Institutional Review Boards (IRBs) or Ethics Committees (ECs)** for human-subjects research.
- **Institutional Ethics Committees (IECs)** under Indian Council of Medical Research (ICMR) guidelines.
- **Research Integrity Offices** handling allegations of misconduct.
- **Funding agency policies** (e.g., DST, SERB, UGC in India) requiring ethics training and certification.

In India, **ICMR's National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017)** is a widely cited reference, applicable even to CS research when it involves human data.

5.3 Retractions and Consequences

When ethical violations are confirmed, consequences may include:

- Retraction of publications.
- Blacklisting by funding agencies.
- Termination of employment.
- Loss of academic degrees.
- Civil or criminal liability (especially in cases involving data protection law violations).

Platforms such as **Retraction Watch** document these cases publicly, serving as both deterrent and educational resource.

6. Practical Case Studies

Case Study 1: The ACM Conference on Fairness, Accountability, and Transparency (FAccT)

A paper claimed that a facial recognition system had a demographic parity gap of 2%, but on closer examination, the authors had used a subset of the dataset favourable to the system. The paper was retracted, and the lead author's institution opened a formal investigation. **Lesson:** Selective reporting and benchmark manipulation constitute falsification.

Case Study 2: A/B Testing Without Consent

A social media company ran an A/B test on 5 million users, manipulating the emotional valence of content in their feeds. Internal researchers raised concerns about psychological harm. The study was published without explicit user consent, raising ethical concerns about autonomy, non-

maleficence, and informed consent. **Lesson:** Even large-scale studies must respect informed consent.

Case Study 3: Ghost Authorship in Industry-Academia Collaboration

A university-industry collaboration produced a paper where the company's medical writer drafted the manuscript, but only academic authors were listed. When the findings were questioned, the missing writer could not be located to verify the analysis. **Lesson:** Ghost authorship compromises accountability and violates ICMJE criteria.

Case Study 4: Undisclosed Conflict in AI Hiring Tool Research

A researcher published a paper praising an AI hiring tool for reducing bias. It later emerged that the researcher held stock options in the company that produced the tool. **Lesson:** Failure to disclose a financial COI undermines the credibility of the research.

7. Summary

Domain	Core Takeaway
Informed Consent	Voluntary, informed, capacitated, and documented agreement is mandatory.
Confidentiality	Protect participant data through technical and procedural safeguards.
Fabrication/Falsification	Honest reporting is non-negotiable; selective reporting is misconduct.
Authorship Norms	Credit must reflect substantial intellectual contribution; use CRediT.
Conflict of Interest	Disclose all relevant interests; manage or recuse when necessary.

These five domains collectively form the operational definition of research integrity. They are interdependent, context-sensitive, and enforceable through institutional mechanisms. For computer science researchers, ethical awareness is not merely a regulatory burden; it is the foundation for producing work that is scientifically valid, socially responsible, and worthy of public trust.

8. Exam Preparation Tips

- 1. Memorize definitions:** Examiners commonly test the definitions of fabrication, falsification, informed consent, authorship criteria, and conflict of interest.

2. **Apply, don't just define:** Be prepared to be given a hypothetical research scenario and asked to identify ethical violations.
 3. **Know the bodies:** Familiarize yourself with the roles of IRB, ICMJE, COPE, ICMR, and Retraction Watch.
 4. **Connect to CS:** Always bring examples back to computer science contexts—HCI, AI, datasets, benchmarking.
 5. **Use structured responses:** For longer questions, organize your answer using the definition, principle, example, and consequence structure.
 6. **Distinguish similar concepts:** Be ready to differentiate confidentiality vs. anonymity, fabrication vs. falsification, and gift vs. ghost authorship.
 7. **Link to regulations:** Referencing GDPR, DPDPA, ICMR, or HIPAA strengthens answers.
 8. **Discuss consequences:** Examiners appreciate awareness of institutional and legal ramifications of misconduct.
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9. Practice Questions

Short Answer Questions

1. Define informed consent. List its four essential components.
2. Differentiate between fabrication and falsification. Provide one CS-relevant example of each.
3. What is the ICMJE criterion for authorship? Why is each of the four criteria important?
4. Explain the difference between anonymization and pseudonymization. Why does the distinction matter for CS research?
5. What is a conflict of interest? List three types with examples.

Long Answer Questions

1. Discuss the ethical issues surrounding the use of publicly available social media data for research. Address informed consent, confidentiality, and applicable regulations.
2. Critically evaluate the criteria for authorship in large multi-institutional machine learning collaborations. How does CRediT taxonomy help resolve ambiguities?
3. A CS researcher at your institute is suspected of manipulating benchmark results in a paper. Outline the institutional process for investigating this allegation and the potential consequences.
4. Compare and contrast the ICMR (India) and GDPR (EU) frameworks in terms of their treatment of informed consent and data protection. Which is more restrictive, and why?

5. Discuss the role of differential privacy in protecting participant confidentiality. What are its limitations?

Scenario-Based Questions

1. **Scenario:** A PhD student wants to scrape user reviews from a public website for a sentiment analysis study. Identify the ethical concerns and propose safeguards.
2. **Scenario:** Three students collaborate on a project. Only the first student writes the paper, but all three are listed as authors. The other two did perform coding and analysis. Evaluate this authorship claim against ICMJE standards.
3. **Scenario:** You are reviewing a paper for an AI conference. You notice that the lead author is a co-founder of the startup whose product is being evaluated. What should you do?
4. **Scenario:** A dataset used in your research contains personal information, but the dataset's terms of use explicitly prohibit re-identification. Discuss the ethical and legal implications.

Higher-Order Thinking Questions

1. To what extent should research ethics differ between academic and industry research?
 2. Can informed consent be meaningfully obtained for large-scale, pre-existing datasets? Justify your position.
 3. Is data fabrication ever justifiable in research? When, if ever?
 4. How should authorship be attributed in projects involving generative AI tools such as large language models?
 5. Should conflict of interest disclosures be made public for all funded research? Argue both sides.
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10. Suggested Further Reading

- **Belmont Report (1979):** Foundational document for research ethics with human subjects.
- **Declaration of Helsinki (1964, last revised 2013):** WMA statement on ethical principles for medical research.
- **ICMR National Ethical Guidelines (2017).**
- **DPDPA (Digital Personal Data Protection Act), India (2023).**
- **GDPR (General Data Protection Regulation), EU (2018).**
- **ICMJE Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals.**
- **COPE (Committee on Publication Ethics) Guidelines.**

- **Office of Research Integrity (ORI) — Introduction to RCR.**
 - **Martín-Martín, A., et al. (2018).** "Ordinary authors and invisible contributors: The need to improve credit attribution in research."
 - **Retraction Watch Database** — <https://retractionwatch.com>
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End of Lecture 7 Notes.